Carl Charbel Fakhir

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Education

Georgia Institute of Technology Dec 2026
Bachelor of Science in Computer Science *GPA: 3.84/4.00*

Experience

NVIDIA — Santa Clara, CA May 2026 – Present
Software Engineering Intern

- **Architecting** a multi-threaded **Rust** rewrite of **NRAS**, the production attestation service guarding confidential AI inference on every **H100**/**H200** GPU across **DGX Cloud**.
- **Hardening** the deployment path with **HashiCorp Vault**, **Cassandra**, and **Envoy** routing across two **GCP** regions behind **AWS** Route 53 + Cloud Armor; fuzz + coverage CI gates block every merge.
- **Implemented** an **AI-orchestrated attestation workflow** verifying GPU firmware measurements **4x faster** (p99: **480ms** to **110ms**), unblocking real-time scaling of confidential inference.

Georgia Tech — Atlanta, GA Aug 2025 – Dec 2025
Undergraduate TA, CS 3001: Computing & Ethics

- **Translated** dense technical material (encryption, ML bias, surveillance, IP law) into plain-language frameworks for **100+** non-CS undergrads through weekly recitations and case-study reviews.

PhantomPeer VPN — Miami, FL Jun 2024 – Aug 2024
Software Engineering Intern

- **Built** a **C++** packet pipeline with **lock-free queues** and **multithreaded** workers, ingesting **50GB+/day** at **sub-100**μ**s** latency.
- **Tuned** async I/O, zero-copy buffers, and kernel-bypass paths, lifting tunnel throughput **18%** and cutting tail latency **120ms**.
- **Profiled** hot paths via **perf** / **eBPF** on **Linux**; redesigned packet batching for cache-line alignment.

Myki Security — Beirut, Lebanon Jun 2023 – Aug 2023
Cybersecurity Engineering Intern

- **Built** an **AES-256-GCM** vault pipeline in **C++** with **constant-time** ops, securing **20K+** daily queries from timing side-channels.
- **Engineered SIMD-accelerated C++**/**Python** validation tools, cutting credential latency **25%** across **1M+** records.
- **Patched 12 critical vulnerabilities** (SQLi, XSS, token-refresh races) at the protocol/auth layer of production.

Projects

Synapse (Low-Latency RAG & Agent Orchestration) Mar 2026 – Present
Founder & Lead Engineer

- **Architected** an in-memory **graph** + **vector RAG** index over **100K+** nodes with **O(log n)** traversal, **sub-50ms p99** via hybrid dense/symbolic retrieval.
- **Built** a **lock-free multi-agent coordinator** over **MCP** at **5x throughput** vs sequential; deployed on **Docker**/**Kubernetes** with **n8n** driving ingest, embedding, and reindex.
- **Engineered CLI**/**plugin** clients with **backpressure** and **FIFO scheduling**, sustaining **1K+** daily QPS under strict SLA bounds.

Pairs Trading Backtester (Statistical Arbitrage) Apr 2026 – May 2026
Quantitative Researcher

- **Screened 63** US equities via **Engle-Granger cointegration** ($p < 0.05$), **ADF** spread tests, and **OU half-life** filter (2–30d) in Python.
- **Built** a dollar-neutral **z-score mean-reversion** strategy ($\pm 2\sigma/\pm 0.5\sigma/\pm 4\sigma$ bands) with OLS hedge, hitting **Sharpe 1.52** and **CAGR 30.8%** on the top pair.
- **Engineered** a **vectorized walk-forward backtest** with **2 bps** per-leg costs, reporting Sharpe, max DD, hit-rate, turnover per pair.

Crypto Arbitrage Scanner (Cross-Exchange) Apr 2026 – May 2026
Quant Developer

- **Built** an async cross-exchange arb detector in **Python (ccxt)** polling **5 venues** at **sub-500ms** with per-venue fault isolation.
- **Modeled net executable PnL** across taker fees, withdrawal costs, and **top-of-book depth caps** to reject unexecutable spreads.
- **Engineered** a real-time ranker over **15+** live order books, surfacing top opportunities by **net bps** every 2s via `rich.live`.

Skills

Languages: C, C++, Python, Rust, Java, SQL, R, MATLAB, Bash, x86-64 asm

ML & Numerics: NumPy, pandas, scikit-learn, PyTorch, embeddings, RAG, MCP, time-series, walk-forward backtesting

Low-Level & Systems: Linux kernel, perf/eBPF, lock-free structures, SIMD, cache-line tuning, async I/O, zero-copy, GPU, AES/ECDSA/SHA, TEE/attestation

Infra: AWS (EC2, S3, Lambda, CloudWatch), GCP (GKE), Docker, Kubernetes, Cassandra, Vault, Envoy, CI/CD